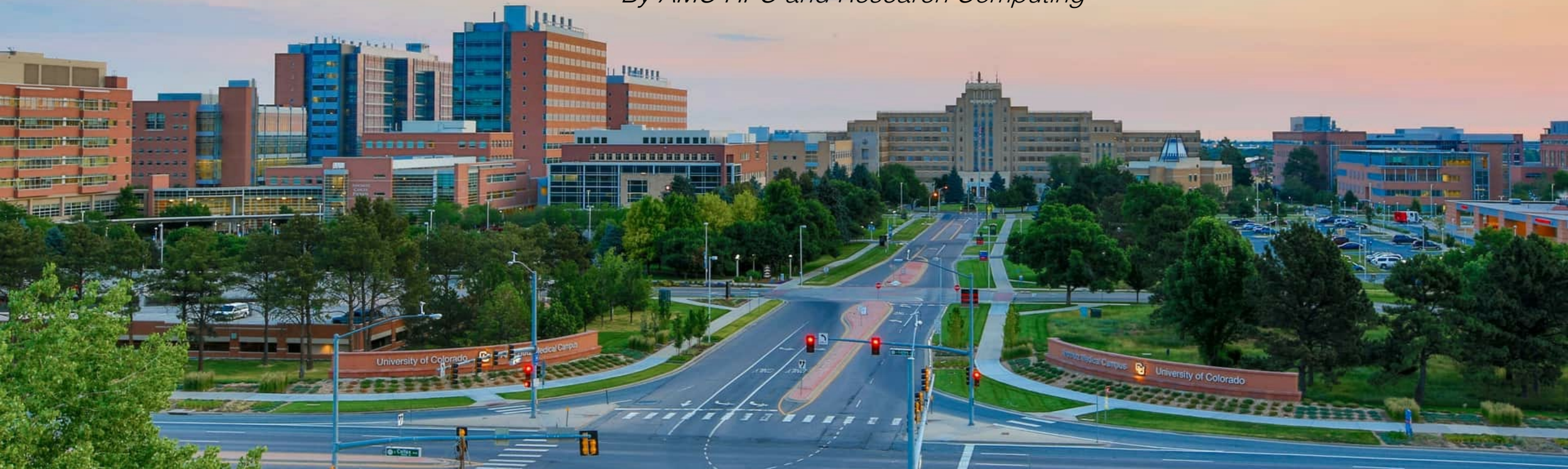




University of Colorado **Anschutz Medical Campus**

The R guide on Alpine

By AMC HPC and Research Computing



R cases on Alpine

There are 3 totally distinct ways to use R on Alpine

- Rstudio on Ondemand with versions **4.4.1** and **4.2.2**. The version **4.4.1** is strongly recommended.
- Under the Alpine shell, there is an R version 4.4.0 that is accessible by using **LMOD**, the modern environment module system for HPC.
- One may decide to install R through **miniforge** or **anaconda**. The latest version they can install is R **4.4.2**



RStudio Server

RStudio Server

This app will launch RStudio Server, an IDE for R on Alpine.

Before utilizing this application, please see the [RStudio Server](#) and [Configuring Open OnDemand interactive applications](#) sections in our documentation. This documentation includes important information regarding quitting an RStudio session. For more information on installing dependencies required by R packages, please see the [installing dependencies for RStudio Server](#) section in our documentation.

RStudio Version

Rstudio 2024.04.2, R 4.4.1

Configuration type

Preset configuration

Preset configuration

4 cores, 4 hours

Launch

* The RStudio Server session data for this session can be accessed under the [data root directory](#).



Currently Loaded Modules:

1) jdk/18.0.1.1 2) R/4.4.0

module load R/4.4.0

module list



module load miniforge



I) Rstudio on Alpine

What is Ondemand Rstudio?

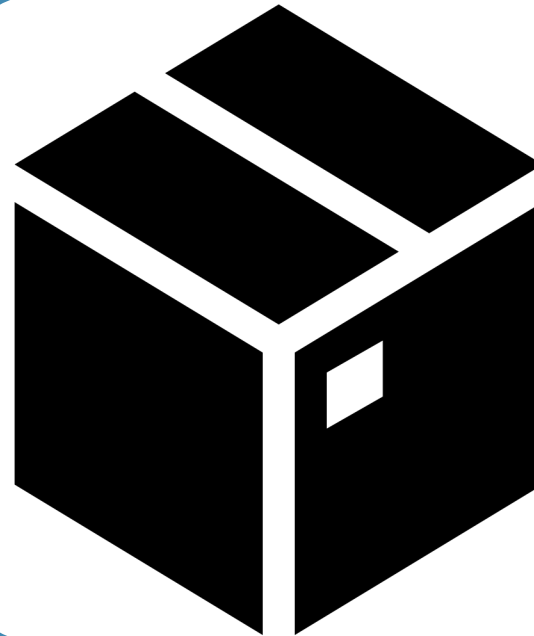
- It was built on top of an apptainer container image, thus, it runs on a different operating system than Alpine.
- A container is a tool that allows you to run different applications that were built on different operating systems on Alpine.
- The Rstudio .sif image is located at
`/curc/sw/containers/open_ondemand/rstudio-server-4.4.1.sif`

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Rstudio



Container

I) Rstudio on Alpine

Use Rstudio on Alpine if:

- If you are used to Rstudio outside of Alpine
- You will need access to a GUI.
- If you think that most of your pipelines will not require more than 16 cores or 60GB of RAM.
- **Note:** * If you are new to Rstudio on Alpine, please refer to this guide: https://github.com/kf-cuanschutz/CU-Anschutz-HPC-documentation/tree/main/Rstudio_related_scripts

* If you wish to run your Rstudio contained environment as a slurm batch script, please refer to this guide: https://github.com/kf-cuanschutz/CU-Anschutz-HPC-documentation/blob/main/Rstudio_on_Slurm_.md

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RStudio Version

Rstudio 2024.04.2, R 4.4.1

Configuration type

Custom configuration

Cluster

Alpine

Account

amc-general

Partition

ahub

QoS

interactive

Time

6

Number of cores

16

Reservation (default is None)

None

gres options (default is None)

None

Launch

* The RStudio Server session data for this session can be accessed under the [data root directory](#).

Rstudio for higher RAM jobs (1)

You may use the acpu partition with Rstudio if:

- You will need access to a GUI.
- Your pipeline will require RAM **higher** than 60G and **lower** than 240G

Request access to the session as shown on the right:

- Since each core allocates around 3.75G, you will need to request at least 17 cores.
- Make sure to select the “acpu” partition and the “cpu-normal” as qos.

RStudio Version
Rstudio 2024.04.2, R 4.4.1

Configuration type
Custom configuration

Cluster
Alpine

Account
amc-general

Partition
acpu

QoS
cpu-normal

Time
1

Number of cores
17

Reservation (default is None)
None

gres (default is None)
None

nodelist (default is None)
None

Rstudio for higher RAM jobs (2)

You may use the amem partition with Rstudio if:

- You will need access to a GUI.
- Your pipeline will require RAM **higher** than 240G.

Request access to the session as shown on the right:

- Since each core allocates either 16G or 21.5G, so you will need to request at least 16 cores.
- Make sure to select the “amem” partition and the “mem-normal” as qos.
- For more information regarding hardware details, please refer to this page:
<https://curc.readthedocs.io/en/latest/clusters/alpine/alpine-hardware.html#university-of-colorado-boulder-contribution>

RStudio Version

Rstudio 2024.04.2, R 4.4.1

Configuration type

Custom configuration

Cluster

Alpine

Account

amc-general

Partition

amem

QoS

mem-normal

Time

1

Number of cores

16

II) LMOD R

What is LMOD R?

- It is accessible through the use of **LMOD**, the modern environment module system for HPC.
- You need to be on a compute node interactively or you need to submit a slurm batch script.
- We only recommend to use the version **4.4.0** to install your packages!!!
- Note that some advance package installation might requires some dependencies. Please open a ticket at hpcsupport@cuanschutz.edu if you are stuck!

```
acompile --ntasks=4 --time=12:00:00
```

Compute
node

```
module avail R
```

Check for
LMOD R

```
-----  
R/3.6.3  
R/4.2.2  
R/4.4.0
```

```
module load R/4.4.0  
which R  
-----  
/curc/sw/install/R/4.4.0/bin/R
```

Load R/4.4.0
and confirm

II) LMOD R

Use LMOD R if:

- You will run heavy job on Alpine, such as high memory computation jobs.
- You will not need to use a GUI interactively.
- You need to run distributed computations, such as with gnu parallel.

```
#!/bin/bash
```

```
#SBATCH --partition=acpu
```

```
#SBATCH --job-name=R_test
```

```
#SBATCH --output=R_test.%j.out
```

```
#SBATCH --error=R_test.%j.err
```

```
#SBATCH --account=amc-general
```

```
#SBATCH --qos=cpu-normal
```

```
#SBATCH --nodes=1
```

```
#SBATCH --ntasks=4
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-type=mail-user=user@cuanschutz.edu
```

```
#SBATCH --time=00:20:00
```

```
# Load the compiler that R depends on.  
module load R/4.4.0
```

```
# Run the R command  
Rscript R_test.█
```



III) Miniforge R

What is miniforge/anaconda R?

- It gives you the possibility to install many R related packages through miniforge.
- Note that your miniforge R located in your miniforge environment will be very distinct from LMOD R!!!



III) Miniforge R

Use miniforge/anaconda R if:

- You are a frequent miniconda/anaconda user.
- If you have followed this guide already:
https://github.com/kf-cuanschutz/CU-Anschutz-HPC-documentation/blob/main/miniforge_migration.md
- If you plan to share the environment you are using with the rest of your lab on Alpine. For example, one can export a miniforge environment as yml.
- Consider this r-irkernel guide if you plan on using Jupyterlab:
https://curc.readthedocs.io/en/latest/open_ondemand/jupyter_session.html

```
acompile --ntasks=4 --time=12:00:00
```

Compute
node

```
module load miniforge
```

Load
miniforge

```
conda create --name r_test_miniforge  
r_test_mini r-base r-essentials
```

Create
ENV





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THANK YOU